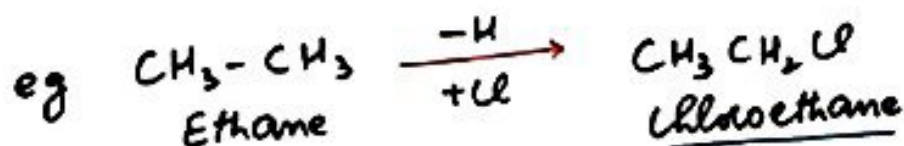


Haloalkanes and Haloarenes

The replacement of hydrogen atom(s) in a hydrocarbon, aliphatic or aromatic by halogen atom(s) result in formation of alkyl halide (Haloalkane) and Aryl halide (Haloarenes)



- In Haloalkane, the halogen atom is bonded to sp^3 hybridized carbon atom
- In Haloarene, the halogen atom is bonded to sp^2 hybridized carbon atom

Halogen containing organic compounds

Natural

- Chlorine containing antibiotic → Chloramphenicol
→ Treatment of typhoid
- Body produce Iodine containing hormone → Thyroxine
→ Deficiency caused Goiter

Synthetic

Chloroquine → Treatment of malaria

Halothane → Anaesthetic during surgery

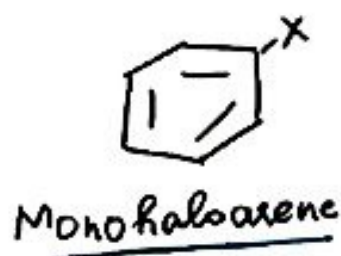
Fully fluorinated compounds → Potential blood substitutes in surgery.

Classification → Haloalkane and haloarene

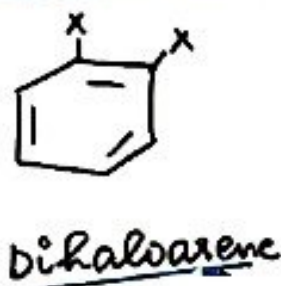
1. On basis of halogen atoms

They are classified as mono, di or poly halogen compounds depending on whether they contain one, two or more halogen atom

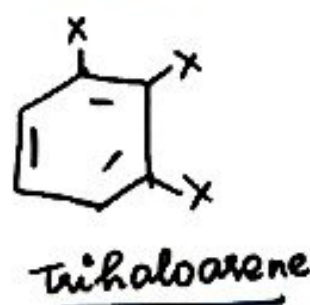
C_2H_5X
Monohaloalkane



$\begin{array}{c} CH_2X \\ | \\ CH_2X \end{array}$
Dihaloalkane



$\begin{array}{c} CH_2X \\ | \\ CHX \\ | \\ CH_2X \end{array}$
Trihaloalkane



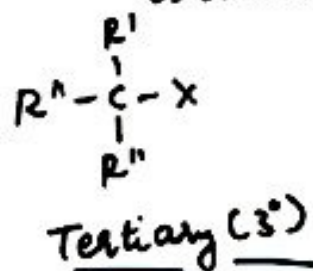
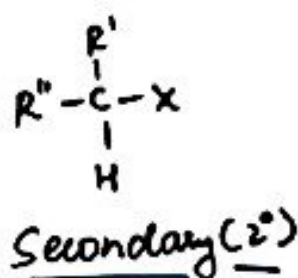
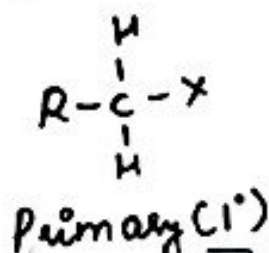
2. On basis of hybridisation of carbon bonded to the halogen

Compound containing $sp^3 C-X$ bond ($X = F, Cl, Br, I$)

① Alkyl halides or haloalkanes ($R-X$)

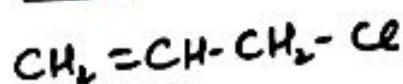
In alkyl halides, the halogen atom is bonded to an alkyl group (R)

- They are further classified as Primary, Secondary or tertiary according to nature of carbon to which halogen is attached.



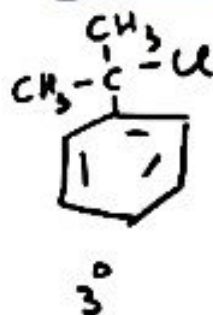
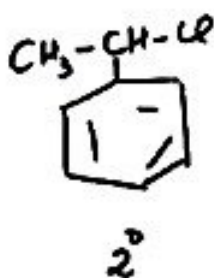
⑤ Allylic halides

These are the compounds in which the halogen atom is bonded to allylic carbon i.e. the sp^3 hybridised carbon next to carbon-carbon double bond.



⑥ Benzylic halides

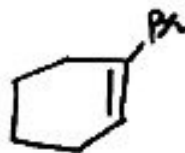
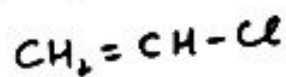
These are the compounds in which halogen atom is bonded to sp^3 hybridised carbon next to aromatic ring.



Compounds containing $sp^2 C-X$ bond

① Vinyl halides

These are the compounds in which the halogen atom is bonded to an sp^2 hybridised carbon atom of Carbon-Carbon double bond ($C=C$)



② Aryl halides

These are the compounds in which halogen atom is bonded to sp^2 hybridised carbon atom of aromatic ring



Dihaloalkane

→ Dihaloalkane having same type of halogen atoms are classified as

1. Geminal halides

→ Halogen atoms are present in same carbon atom.

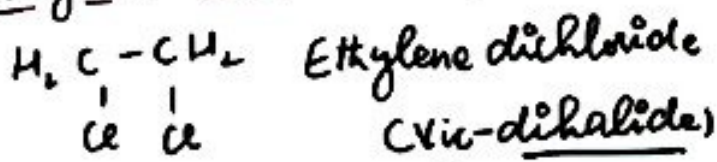
→ Common name → alkylidene halide or alkylidene dihalide



2. Vicinal halide

→ Halogen atoms are present on the adjacent carbon atoms

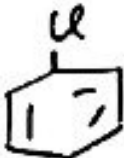
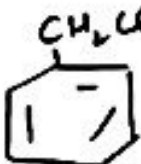
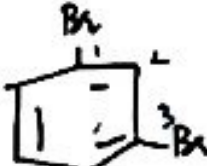
→ Common name → Alkylene halide or alkylene dihalide



Nomenclature

Rule → Functional group > No of substituents > Lowest set of locant
(Double or triple bond) > Alphabetical order

		<u>Haloalkanes</u>	
<u>IUPAC</u>	$\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{Br}$	$\text{CH}_3 - \underset{\text{Cl}}{\text{CH}} - \text{CH}_3$	$\text{CH}_3 - \text{CH}_2 - \underset{\text{Cl}}{\text{CH}} - \text{CH}_3$
	1-Bromopropane	2-Chloropropane	2-Chlorobutane
<u>Common name</u>	n-propyl bromide	Isopropyl chloride	sec-butyl chloride
	$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3 - \text{C} - \text{Br} \\ \\ \text{CH}_3 \end{array}$	$\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_2 - \text{Br}$	$\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{Cl}$
<u>IUPAC</u>	2-Bromo-2-Methylpropane	1-Bromo-2-methylpropane	1-chlorobutane
	tert-butyl bromide	Isobutyl bromide	n-butyl chloride

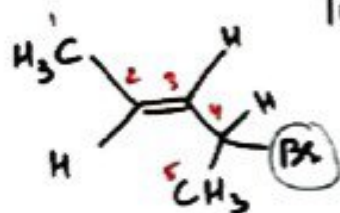
	CHCl_3	CHBr_3	CHI_3	CCl_4
<u>IUPAC</u>	Trichloro methane	Tri bromo methane	Tri iodo methane	Tetra chloro methane
<u>Common name</u>	<u>Chloroform</u>	<u>Bromoform</u>	<u>Iodoform</u>	Carbon tetra chloride
				
	Chlorobenzene	Benzyl chloride	Benzal dichloride	Benzo trichloride
				
	1,2-Dibromobenzene o-Dibromobenzene	1,3-Dibromobenzene m-Dibromobenzene	1,4-Dibromobenzene p-Dibromobenzene	1,3,5-Tri bromobenzene Sym-tri bromobenzene

	CHCl_3	CHBr_3	CHI_3	CCl_4
<u>IUPAC</u>	Trichloro methane	Tri bromo methane	Tri iodo methane	Tetra chloro methane
<u>Common name</u>	<u>Chloroform</u>	<u>Bromoform</u>	<u>Iodoform</u>	Carbon tetra chloride
				
	Chlorobenzene	Benzyl chloride	Benzal dichloride	Benzo trichloride
				
	1,2-Dibromobenzene o-Dibromobenzene	1,3-Dibromobenzene m-Dibromobenzene	1,4-Dibromobenzene p-Dibromobenzene	1,3,5-Tri bromobenzene Sym-tri bromobenzene

Example → IUPAC name

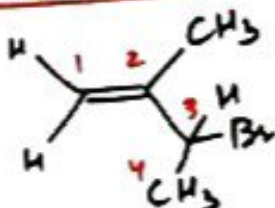
10.2

(i)



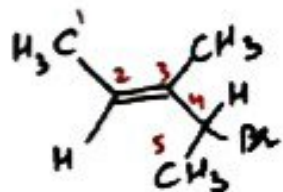
4-Bromopent-2-ene

(ii)



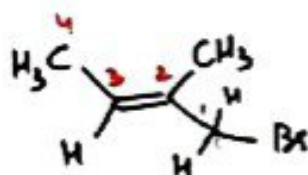
3-Bromo-2-methyl
but-1-ene

(iii)



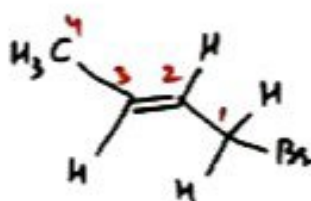
4-Bromo-3-methyl
pent-2-ene

(iv)



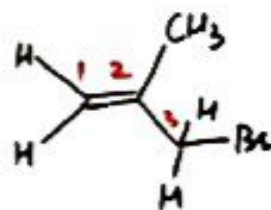
1-Bromo-2-methyl
but-2-ene

(v)



1-Bromobut-2-ene

(vi)

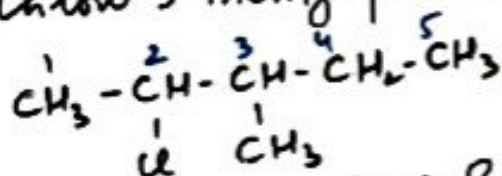


3-Bromo-2-methyl
propene

Intext Question 10.1

Write structure of the following compounds

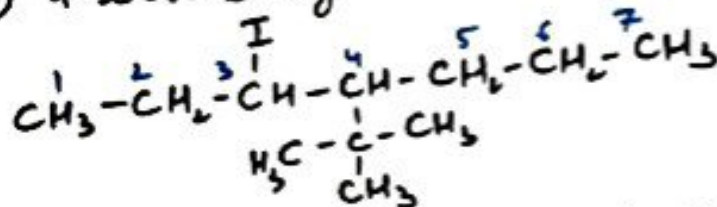
(1) 2-chloro-3-methylpentane



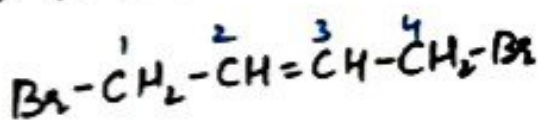
(2) 1-chloro-4-ethylcyclohexane



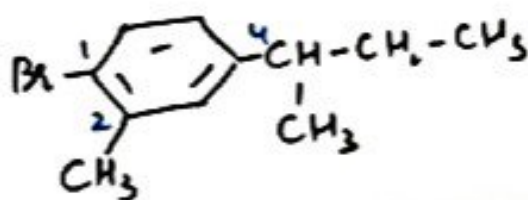
(3) 4-tert-butyl-3-iodoheptane



(4) 1,4-Dibromobut-2-ene

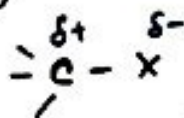


(5) 1-bromo-4-sec-butyl-2-methylbenzene



Nature of C-X Bond

- Halogen atoms are more electronegative than carbon.
- Bond of alkyl halide is polarised



- Size of halogen atom increases as we move down the group
- Carbon halogen bond length also increases.

Bond length $\text{CH}_3\text{F} < \text{CH}_3\text{Cl} < \text{CH}_3\text{Br} < \text{CH}_3\text{I}$

Bond enthalpy $\text{CH}_3\text{F} > \text{CH}_3\text{Cl} > \text{CH}_3\text{Br} > \text{CH}_3\text{I}$

Dipole moment $\text{CH}_3\text{Cl} > \text{CH}_3\text{F} > \text{CH}_3\text{Br} > \text{CH}_3\text{I}$

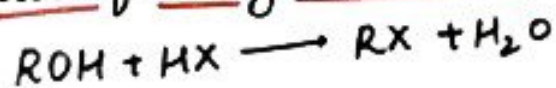
$$\mu = \underbrace{q \times r}_{\text{Charge}} \rightarrow \text{Distance}$$

Methods of Preparation of Haloalkanes

1. From alcohol

- Alcohol can be converted to haloalkane by substitution of -OH group with halogen atom.

① By reaction of halogen acid (HX)



① Chloroalkane (RCl)

- Chloroalkane is obtained by treating alcohol with HCl in presence of anhydrous zinc chloride



Anhydrous ZnCl_2 act as dehydrating agent and thus favour the forward reaction (Le Chatelier principle)

② Bromoalkane (RBr)

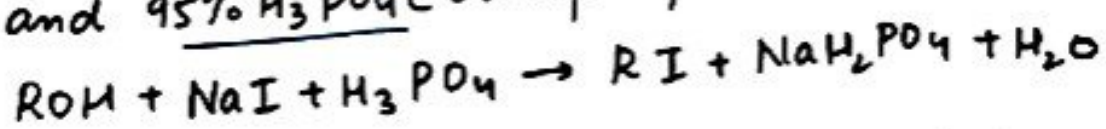
It is obtained by heating alcohol with KBr or NaBr and conc H₂SO₄. HBr is generated in situ (during the reaction)



→ Reaction can be carried out by conc. solution of HBr (48%)

③ Iodoalkane (RI)

It is obtained by heating alcohol with KI or NaI and 95% H₃PO₄ (orthophosphoric acid)

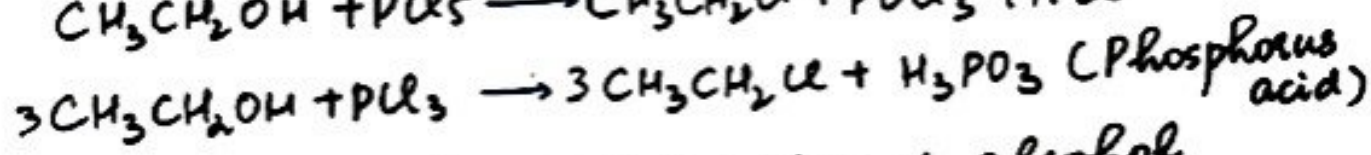
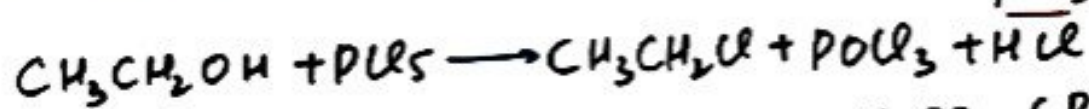


Reaction of halogen acid → HI > HBr > HCl

Reactivity of alcohol with HX → 3° > 2° > 1°

⑥ Reaction with Phosphorus halide

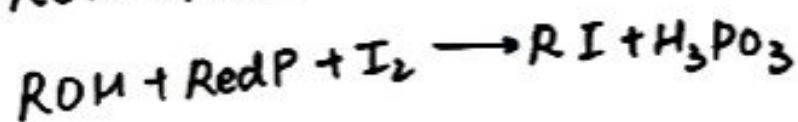
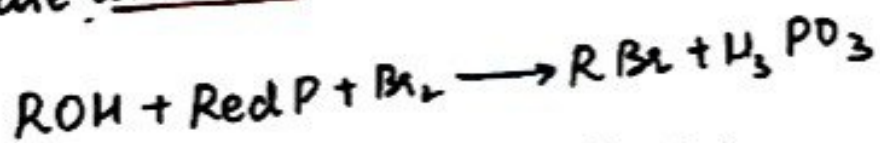
Chloroalkane are obtained by reaction of alcohol with PCl₃ or PCl₅



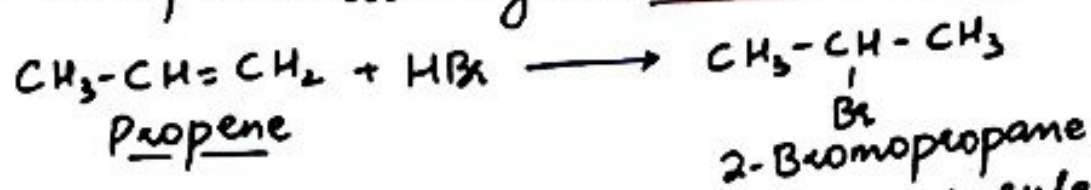
→ Bromo and Iodoalkane → Reaction of alcohol with mixture of Red phosphorus and Br₂ or I₂

→ PBr₃ or PI₃ are prepared insitu

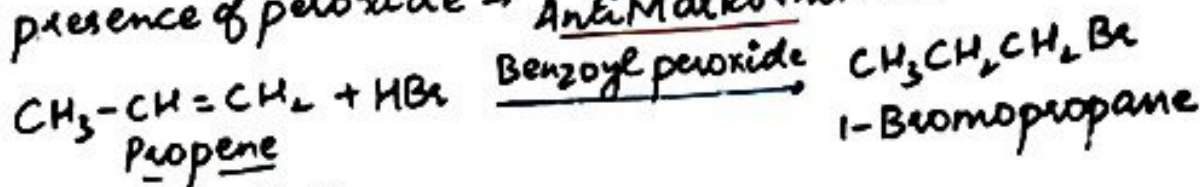
→ They are unstable and cannot be stored for longer period



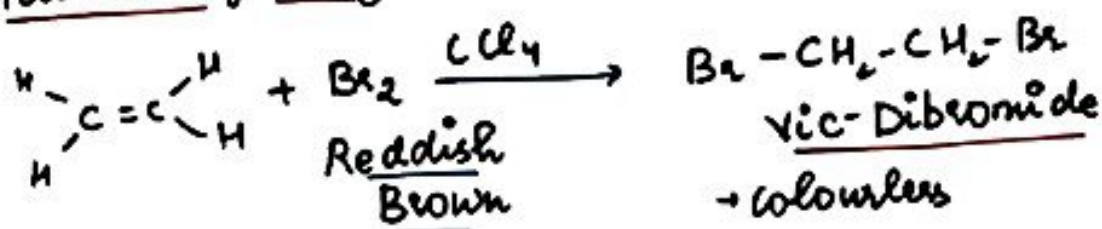
In unsymmetrical alkene the addition of halogen acid take place according to Markovnikov's rule



In presence of peroxide $\rightarrow$ AntiMarkovnikov's rule

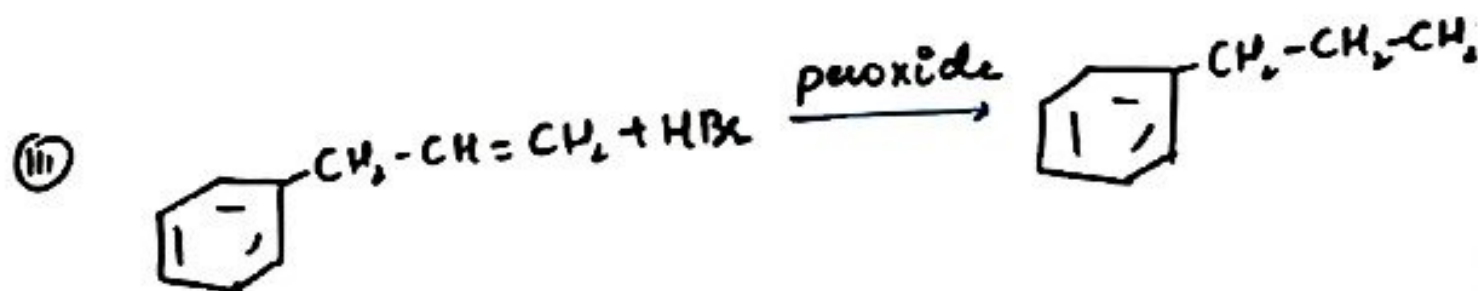
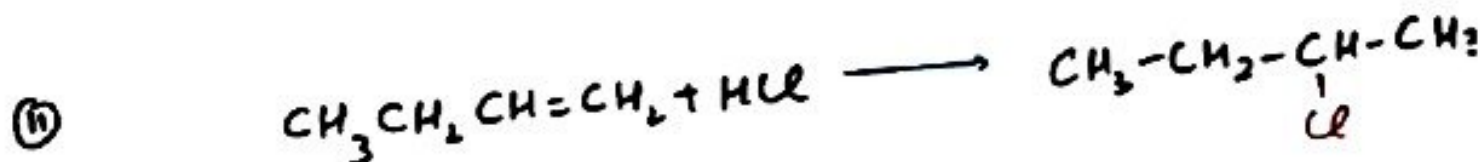
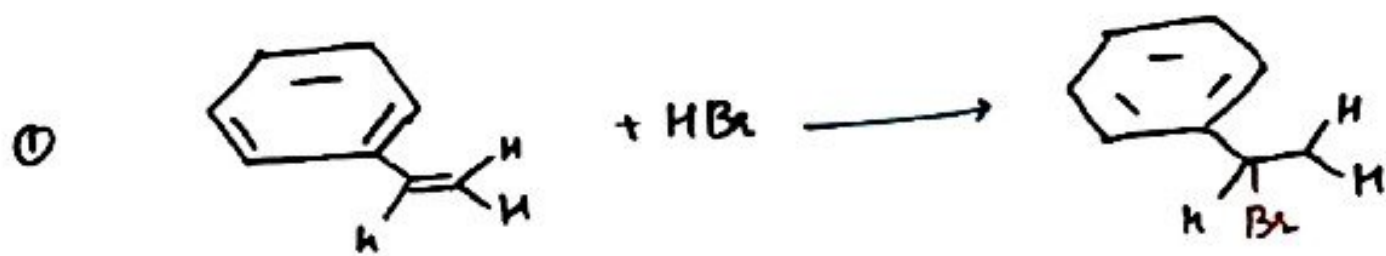


(ii) Addition of halogen



Discharge of reddish brown colour $\rightarrow$ Test for unsaturation

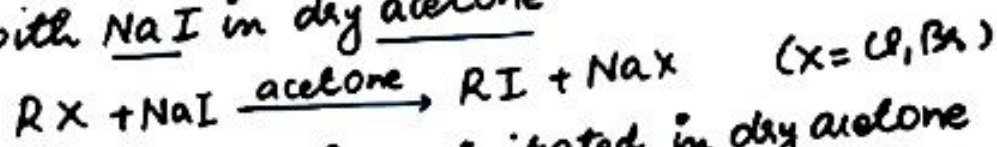
Example 10.4 Write product of following reaction



3. Halogen exchange

① Finkelstein reaction

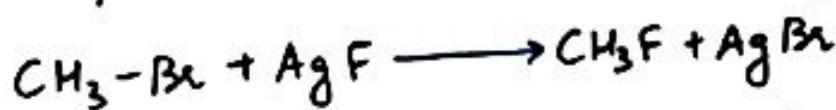
Alkyl iodide can be prepared by reaction of alkyl chloride/bromide with NaI in dry acetone



- NaCl or NaBr formed is precipitated in dry acetone
- It facilitates the forward reaction by Le Chatelier's principle

② Swarts reaction

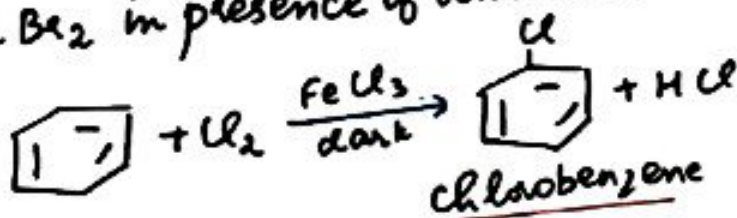
→ Alkyl fluoride can be obtained by heating alkyl chloride/bromide in presence of metallic fluoride such as AgF, Hg₂F₂, CoF₂ or SbF₃



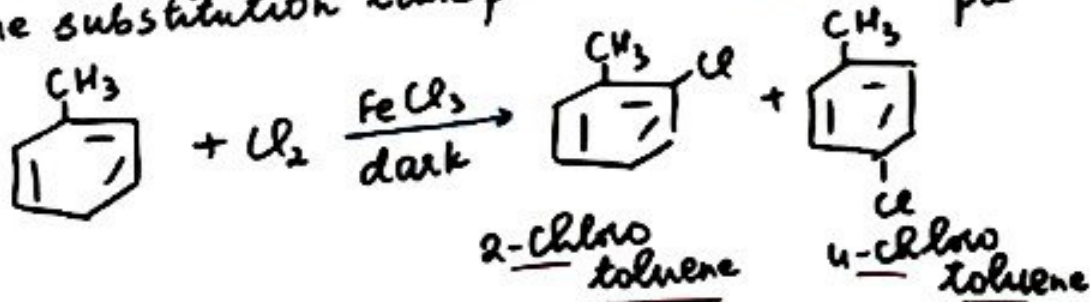
Preparation of Haloarenes

1. From arenes (by electrophilic substitution)

- Aryl chloride and bromide can be easily prepared
- Cl₂ and Br₂ in presence of Lewis acid catalyst like Iron or Iron(III) chloride



→ In toluene substitution take place at ortho and para positions



Ortho and para isomers can be easily separated due to large difference in melting point

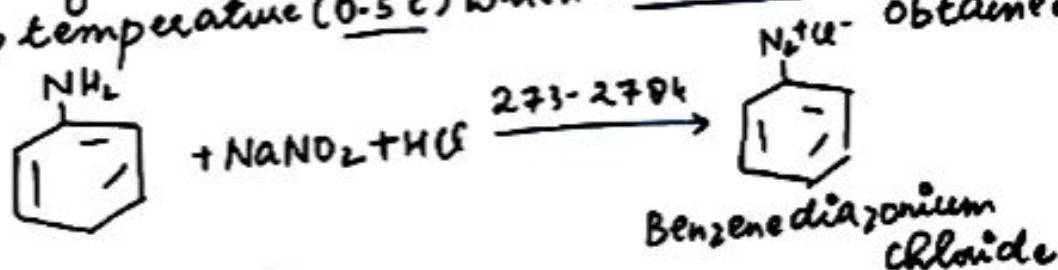
→ Fluoro compounds are not prepared by this method due to high reactivity of Fluorine

→ Reaction with iodine are reversible in nature and require presence of an oxidising agent (HNO_3 , HIO_4) to oxidise HI to I_2

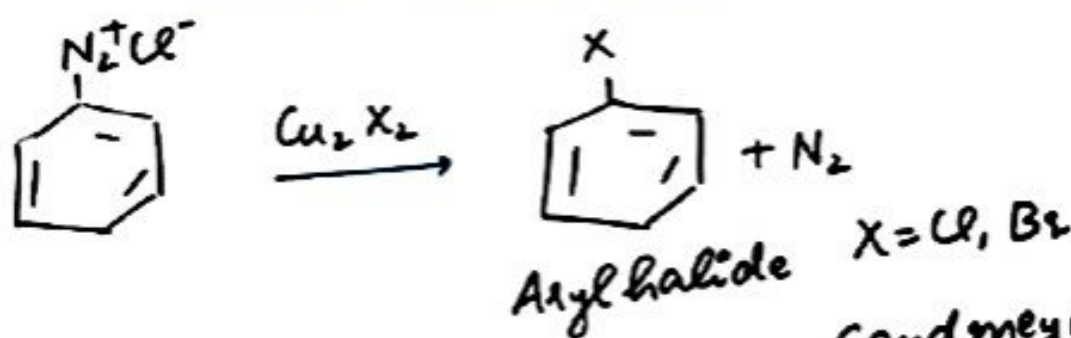


2. From diazonium compound (By Sandmeyer's reaction)

→ Primary aromatic amine react with nitrous acid (NaN + HO) at low temperature ($0-5^\circ\text{C}$) when diazonium salt is obtained



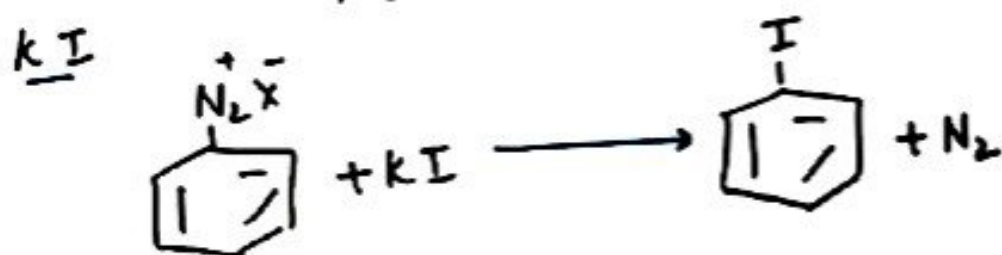
→ This reaction is known as Diazotisation



Sandmeyer reaction

→ Replacement of diazonium group by iodine does not require presence of cuprous halide

→ It is done simply by shaking diazonium salt with

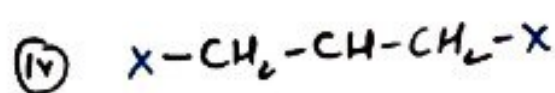
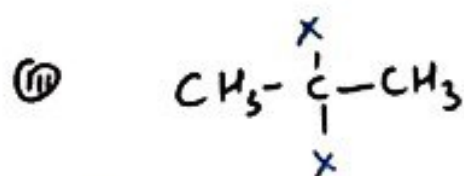
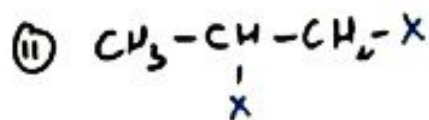
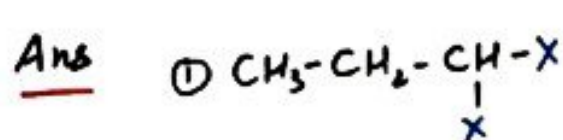


Intext Questions

10.2 Why is sulphuric acid not used during the reaction of alcohols with KI?

Ans H_2SO_4 cannot be used along with KI in the conversion of an alcohol to an alkyl iodide as it converts KI to corresponding HI and then oxidizes HI to I_2 .

10.3 Write structure of different dihalogen derivative of propane.

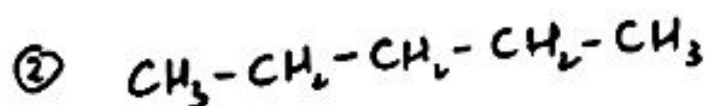
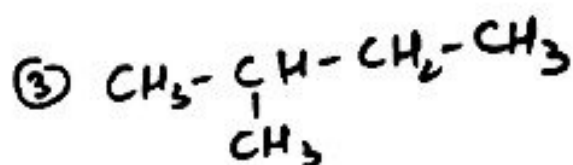
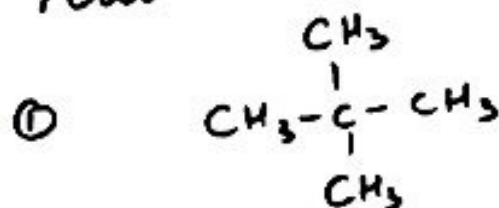


Among the isomeric alkane of molecular formula C_5H_{12} . Identify one that on photochemical chlorination yields

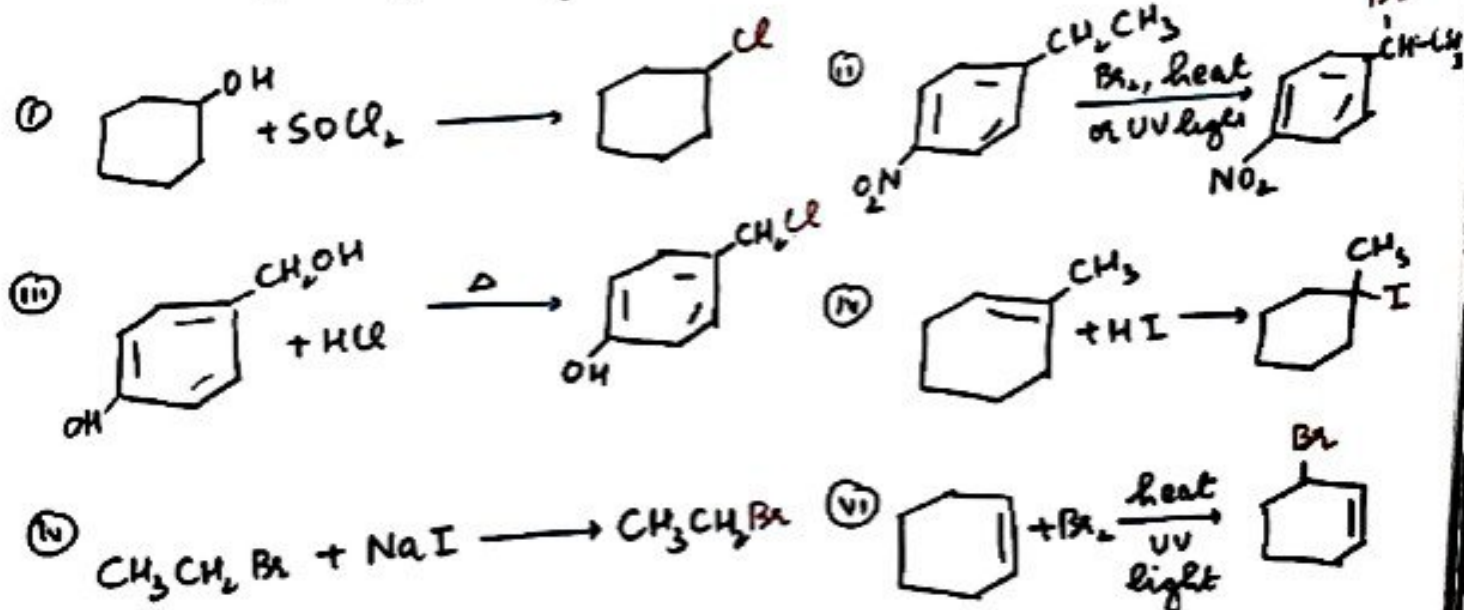
① Single monochloride

② Three isomeric monochlorides.

③ Four isomeric monochlorides



10.5 Draw the structure of major monohalo product in each of the following reactions



Physical properties

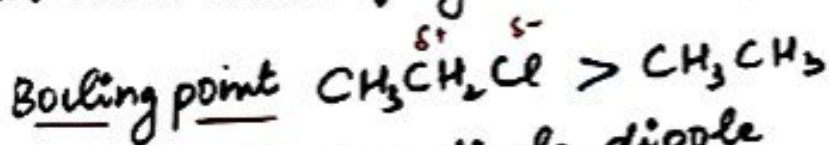
- Pure haloalkanes are colourless
- However Bromoalkane and Iodoalkane develop colour when exposed to light and stored for longer period.

Melting and Boiling point

- Methyl chloride, methyl bromide, ethyl chloride and some chlorofluoromethane are gases at room temperature
- Higher members are liquid or solid.
- Boiling point → $RF < RCl < RBr < RI$
- With increase in size and mass of halogen atom the magnitude of vanderwaal force of attraction increases

$$[B.P \propto \text{Molecular mass}]$$

→ Boiling point of chloride, bromide and Iodide are higher than that of hydrocarbon of comparable Molecular mass



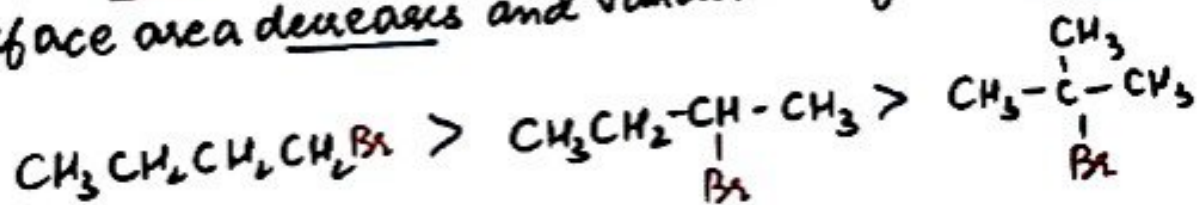
Greater polarity → dipole-dipole

Higher molecular mass → Vanderwaal forces

→ Intermolecular force of attraction are stronger in halogen derivatives

→ Boiling point of isomeric haloalkane decrease with increase in branching.

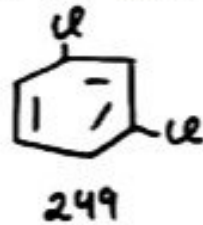
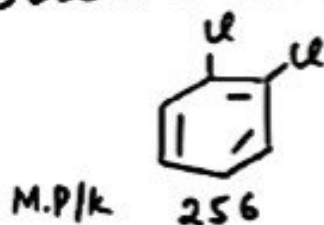
→ Surface area decreases and vanderwaal forces decrease.



→ Boiling point of isomeric dihalobenzenes are very nearly the same.

→ Para isomers have high melting point as compared to their ortho and meta isomers

→ Due to symmetry of para-isomers that fits in crystal lattice better as compared to ortho and meta isomers

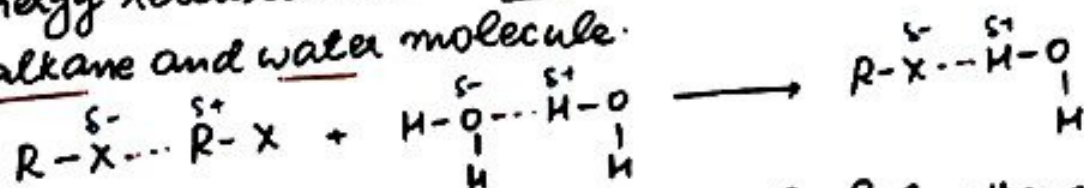


Density → Bromo, iodo and polychloro derivative of hydrocarbon are heavier than water

→ Density increase with increase in number of carbon atom, halogen atom and atomic mass of halogen atom.

Solubility

- Although polar in nature alkyl halides are insoluble in water
- The energy required to overcome the force of attraction between the haloalkane molecules and that required to break hydrogen bonds between water molecules is more than the energy released when new attraction are set up between haloalkane and water molecule.



- Soluble in organic solvent such as alcohol, ether and benzene
- New attraction have ^{same} strength as the one required to separate haloalkane and solvent molecule.

Intext Question

6 Arrange each set of compounds in order of increasing boiling points

① Bromomethane, Bromoform, Chloromethane, Dibromomethane

② 1-chloropropane, Isopropyl chloride, 1-chlorobutane

Text Question

10.6 Arrange each set of compounds in order of increasing boiling points

① Bromomethane, Bromoform, Chloromethane, Dibromomethane

② 1-chloropropane, Isopropyl chloride, 1-chlorobutane

Ans ① Chloromethane CH_3Cl < Bromomethane CH_3Br < Dibromomethane CH_2Br_2 < Bromoform CHBr_3

② Isopropyl chloride $\text{CH}_3-\underset{\text{Cl}}{\text{CH}}-\text{CH}_3$ < 1-chloropropane $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{Cl}$ < 1-chlorobutane $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{Cl}$

Chemical reactions

Reactions of haloalkanes

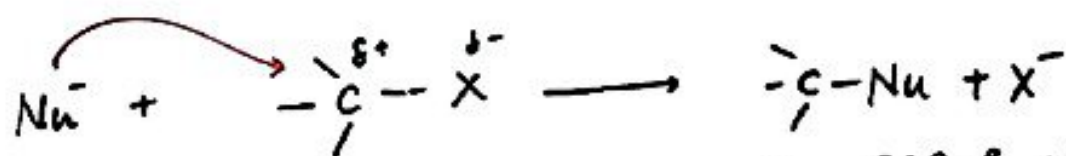
① Nucleophilic substitution

② Elimination reaction

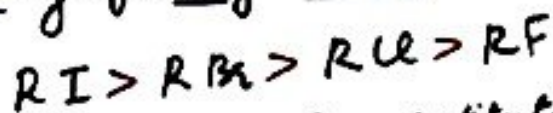
③ Reaction with metal

① Nucleophilic substitution reaction

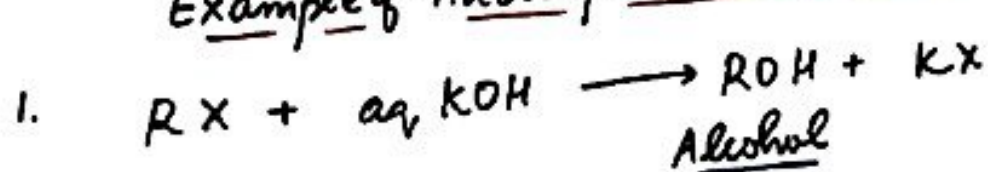
- In haloalkanes, the halogen atom is attached to carbon atom
- Due to presence of partial positive charge on the carbon atom, the nucleophile can attack on electron deficient carbon thereby resulting in the displacement of weak nucleophile, the halide ion (leaving group)
- Nucleophilic substitution reaction



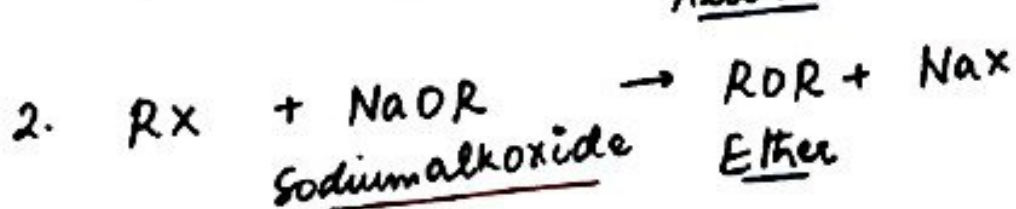
→ Halide ion is substituted only if nucleophile is stronger than it
order of reactivity of alkyl halide



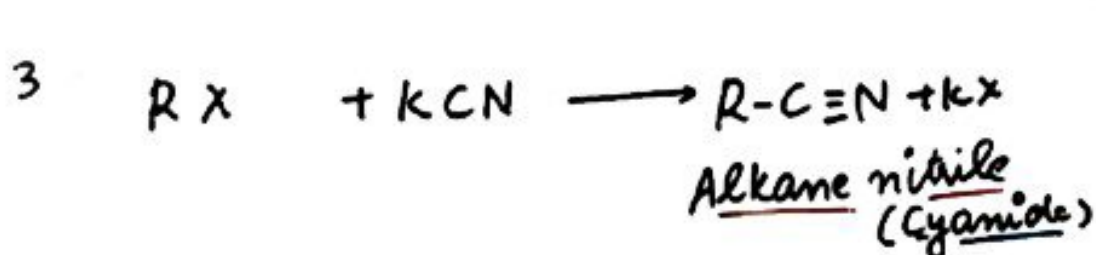
Example of nucleophilic substitution



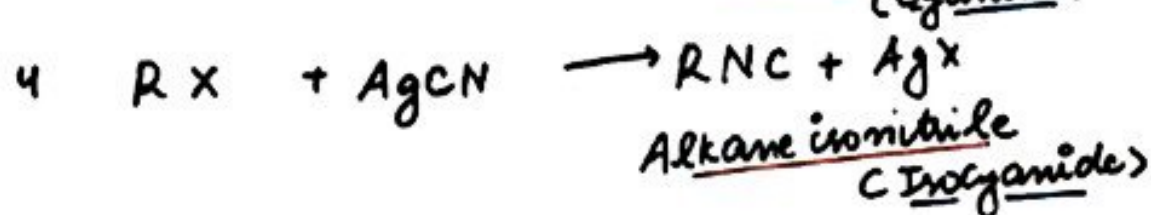
Nucleophile
 OH^-



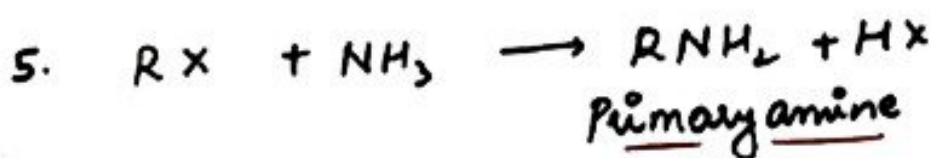
RO^-
Alkoxide



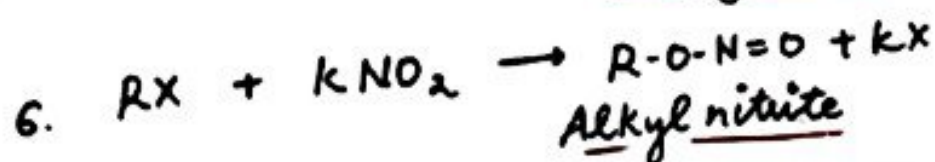
Nucleophile
 $\text{C}\equiv\text{N:}^-$



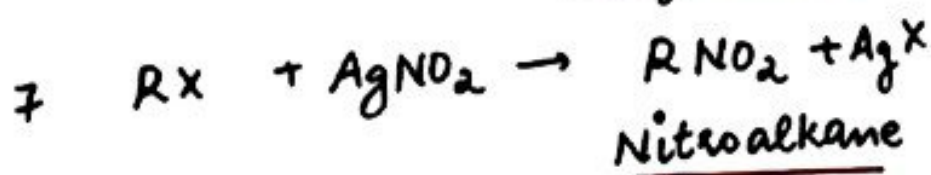
AgCN



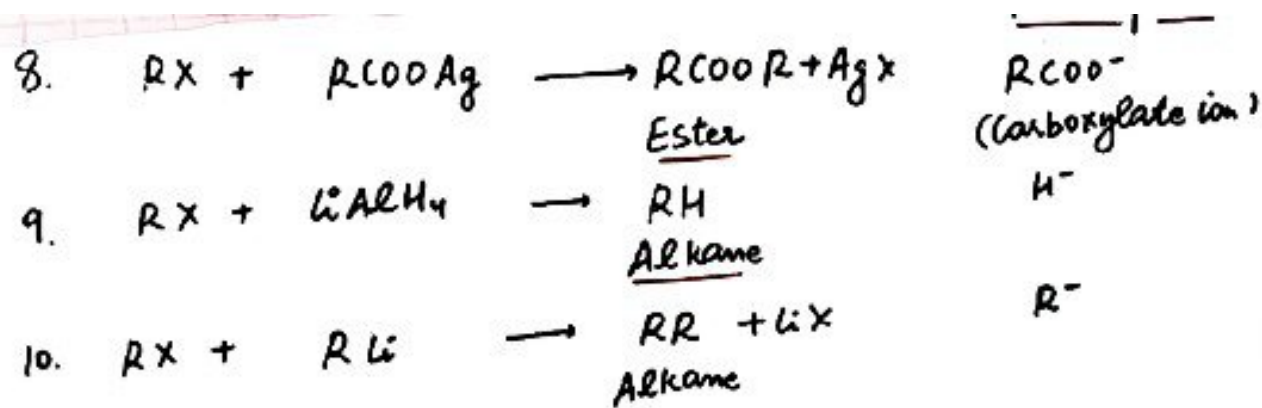
NH_3



O=N-O^-



AgNO_2

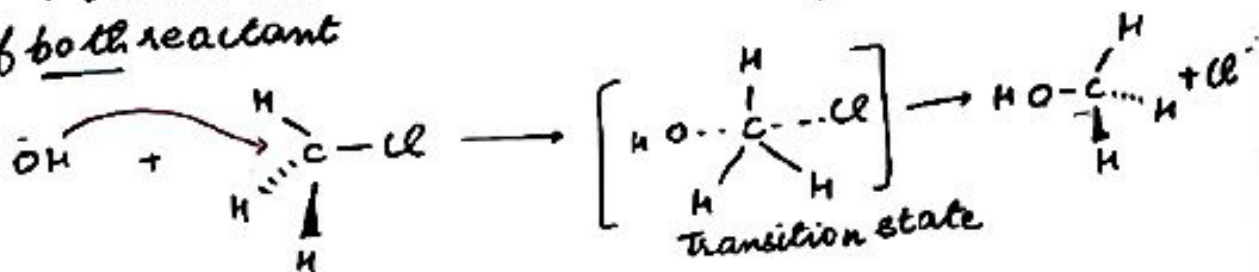


- Ambident nucleophile
- Nucleophile such as cyanide ($:C \equiv N:$) and nitrite ion ($O=N-O^-$) can attack nucleophilic centre from two sites
 - Cyanide → Alkyl cyanoide and alkyl isocyanide
 - Nitrite ion → Alkyl nitrite and nitroalkane

Mechanism

1. Bimolecular nucleophilic substitution (SN^2)

- Follows second order kinetics i.e depend on concentration of both reactant



$$Rate = k [OH^-] [CH_3Cl]$$

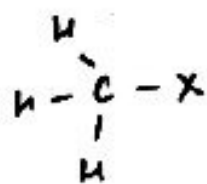
- Incoming nucleophile interact with alkyl halide causing the carbon halide bond to break while forming new carbon-OH bond.

- These two processes take place simultaneously in a single step and no intermediate is formed.

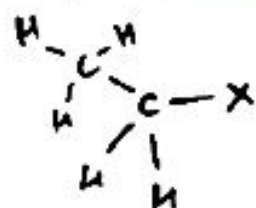
- Inversion of configuration

- Transition state, carbon is simultaneously bonded to five atoms. Such structure are unstable and cannot be isolated.

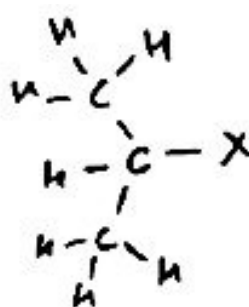
Order of reactivity



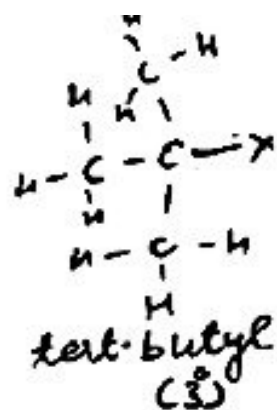
Methyl



Ethyl (1°)



Isopropyl (2°)

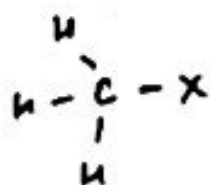


tert-butyl (3°)

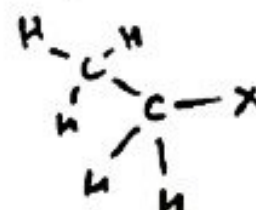
- Presence of bulky substituent on or near the carbon atom have dramatic inhibiting effect.
- Methyl halide react most rapidly because of only three small hydrogen atom
- Tertiary halides are the least reactive because bulky groups hinder the approaching nucleophile.

Primary halide > Secondary halide > Tertiary halide

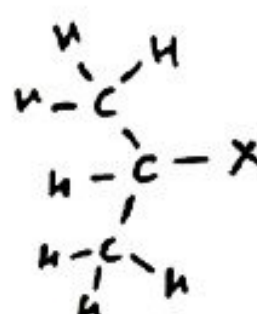
Order of reactivity



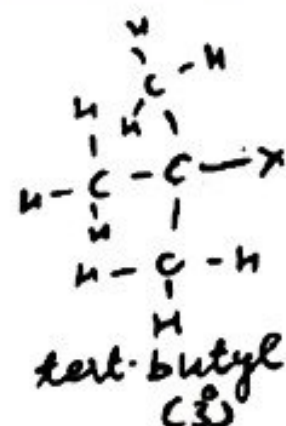
Methyl



Ethyl (1°)



Isopropyl (2°)



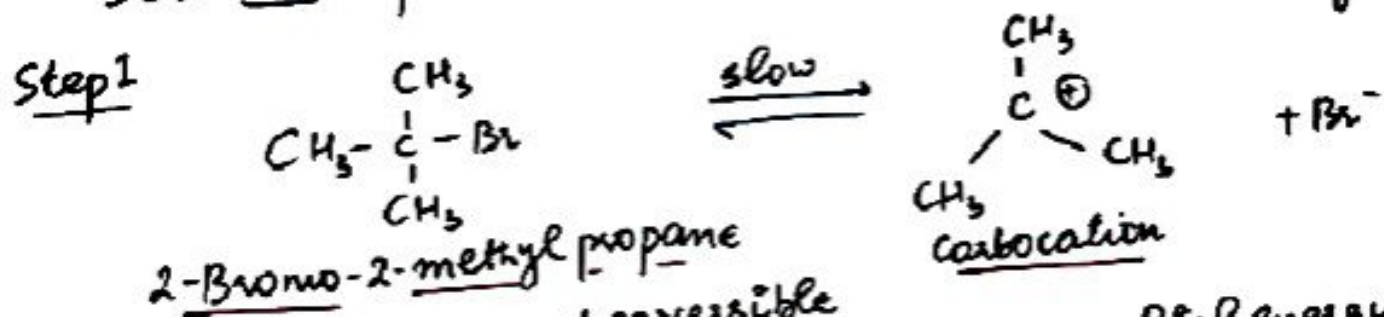
tert-butyl (3°)

- Presence of bulky substituent on or near the carbon atom have dramatic inhibiting effect.
- Methyl halide react most rapidly because of only three small hydrogen atom
- Tertiary halides are the least reactive because bulky group hinder the approaching nucleophile.

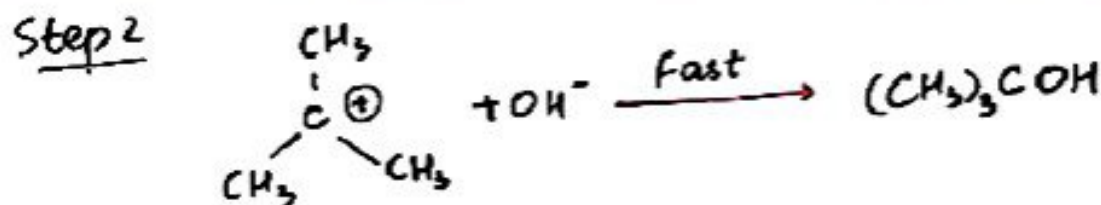
Primary halide > Secondary halide > Tertiary halide

Substitution nucleophilic unimolecular (S_N1)

- S_N1 carried out in polar protic solvent (water, alcohol, acetic acid)
- Follow first order kinetics
- It is two step reaction and carbocation intermediate is formed.

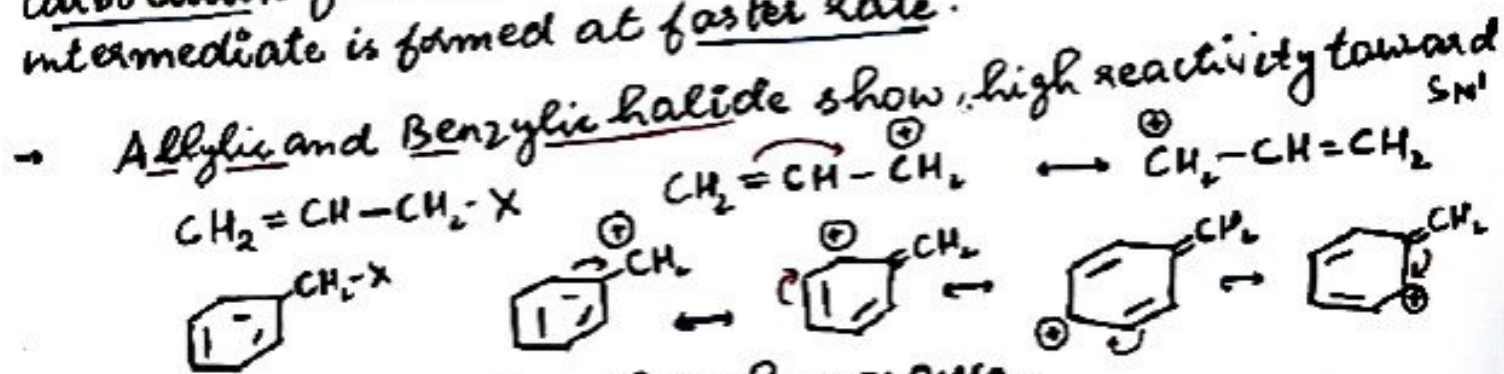


- Step 1 is slow and reversible
 - It involve the C-Br bond breaking for which energy is obtained through solvation of halide ion with proton of protic solvent
 - Rate of reaction depend on slowest step.
- Rate = k [Alkyl halide]



Order of reactivity 3° > 2° > 1°

- 3° alkyl halides are most reactive because the intermediate carbocation formed is most stable. The more stable intermediate is formed at faster rate.

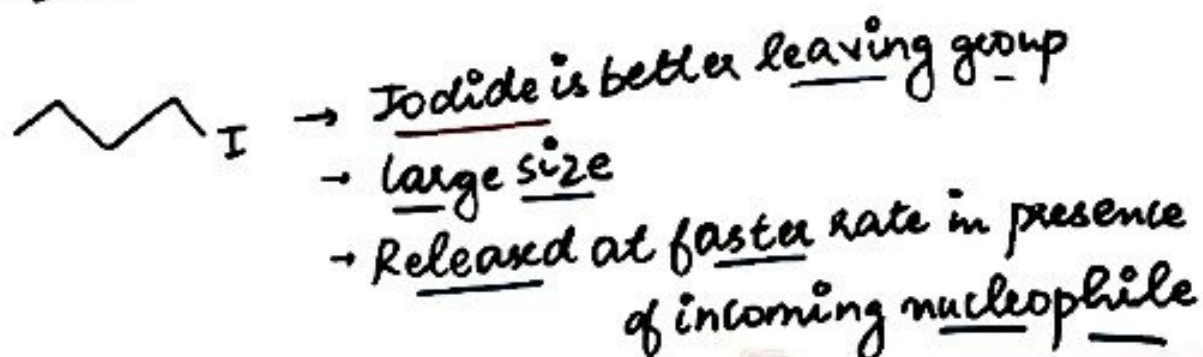
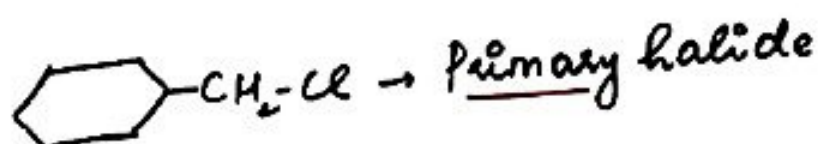
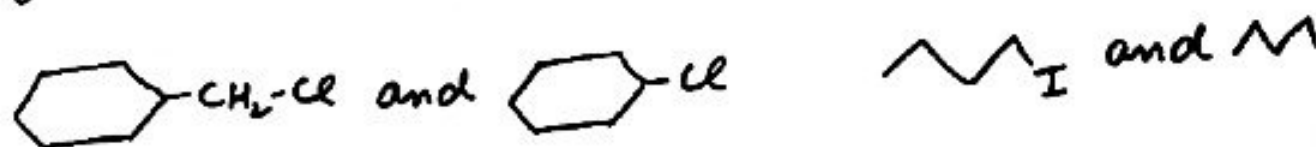


Carbocation stabilize through resonance.

Both mechanism → R-I > R-Br > R-Cl > R-F

Example 10.6

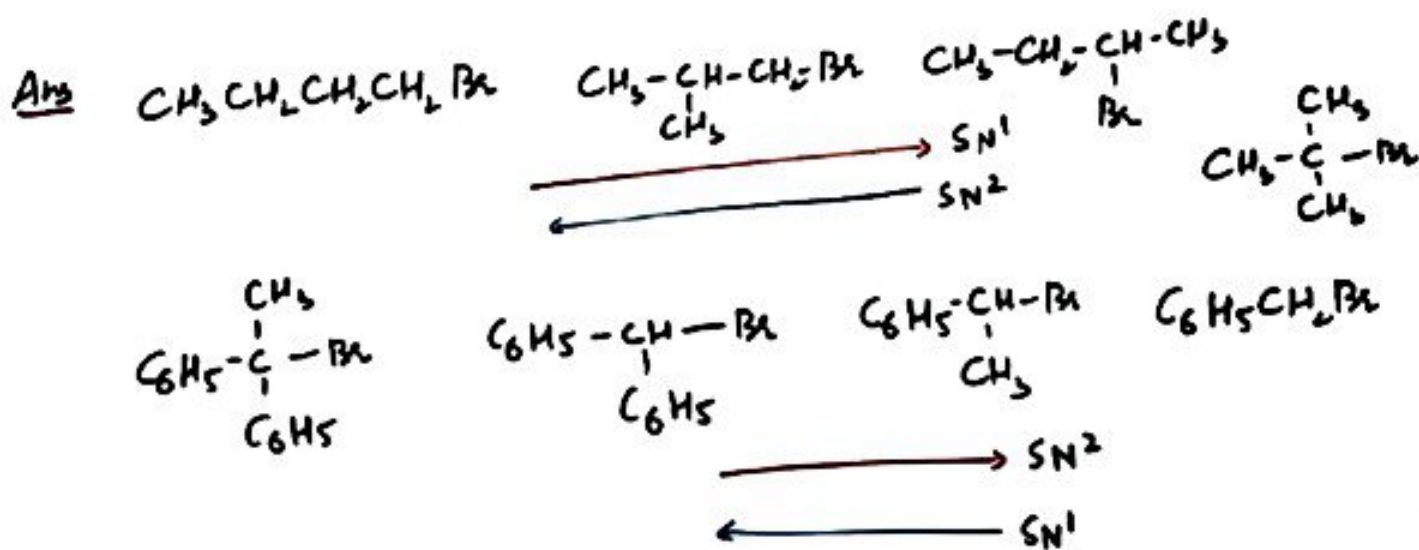
In the following pairs of halogen compounds, which undergo S_N2 reaction faster?



10.7 Predict the order of the reactivity of the following compounds in S_N1 and S_N2 reaction

① The four isomeric bromobutane

② C6H5CH2Br, C6H5CH(C6H5)Br, C6H5CH(CH3)Br, C6H5C(CH3)2(C6H5)Br



① Stereochemical aspects of nucleophilic substitution reaction

① Plane polarized light

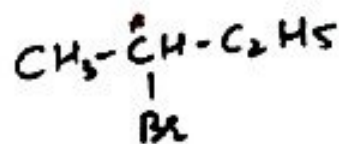
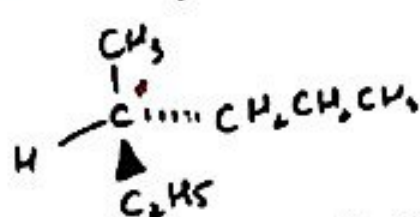
- Under normal condition, the light wave oscillate in infinite number of plane.
- Plane polarised light is a light whose vibration takes place only one of the possible plane.
- Ordinary light can be turned into plane polarized light by passing it through Nicol prism.

② Optical rotation and optically active substance

- When plane polarized light is passed through the liquid or dissolved form of certain substance.
- The plane of oscillation gets rotated through some angle toward left or right of the original oscillation.
- The substance which rotate the plane of polarized light → Optically active substance
 - The substance which rotate the plane of polarized light in the clockwise direction or toward right → dextrorotatory substance
 - d or (+) sign before name of the substance
 - The substance which rotate the plane of polarized light in the anticlockwise direction or toward left → laevorotatory substance
 - l or (-) sign before name of the substance
- The extent of optical rotation caused by the optically active substance is measured by the instrument called polarimeter.

Chirality centre

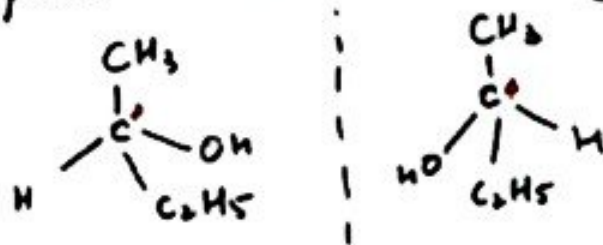
- sp^3 hybridised carbon attached with 4 different substituent is known as chiral carbon or asymmetric carbon or Stereocentre
- The chiral carbon is represented by asterisk ($*$)



- The resultant molecule lack symmetry and is referred to as asymmetric molecule
- The asymmetry of the molecule is responsible for the optical activity in such organic compound.

Chiral molecules

- A molecule which does not have a plane of symmetry is called chiral molecule
- Non-superimposable on its mirror image
- The molecule which contain one chirality centre are non superimposable on their mirror image → chiral



Enantiomers

- Stereoisomers with non superimposable mirror image
- They rotate the plane polarised light through same angle in opposite direction

→ Thus if one enantiomer is dextrorotatory, the other would be laevo rotatory.

→ Enantiomers possess identical physical properties namely melting point, boiling point, solubility, refractive index.

→ They differ with respect to rotation of plane polarized light.

Racemic mixture

→ A mixture containing two enantiomers in equal proportion will have zero optical rotation.

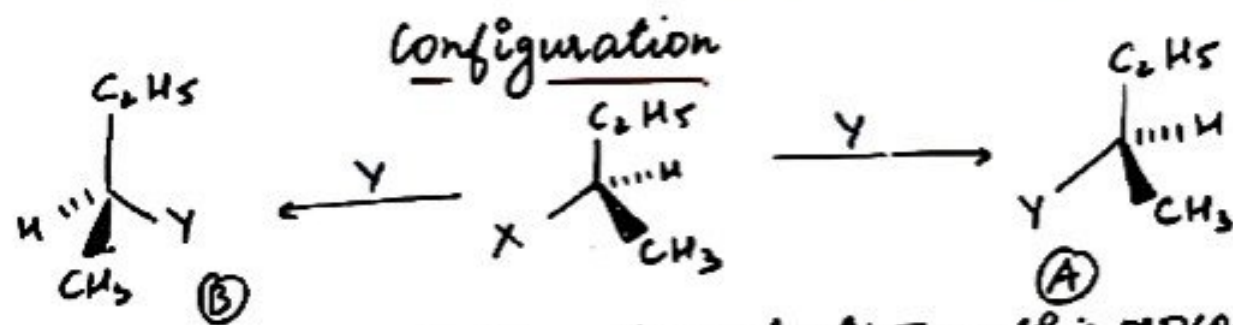
→ The rotation of one isomer will be cancelled by the rotation of other isomer.

→ Such a mixture is called racemic mixture.

→ It is represented by dl or (±)

eg (±) Butan-2-ol

→ The process of conversion of enantiomer into a racemic mixture → Racemisation



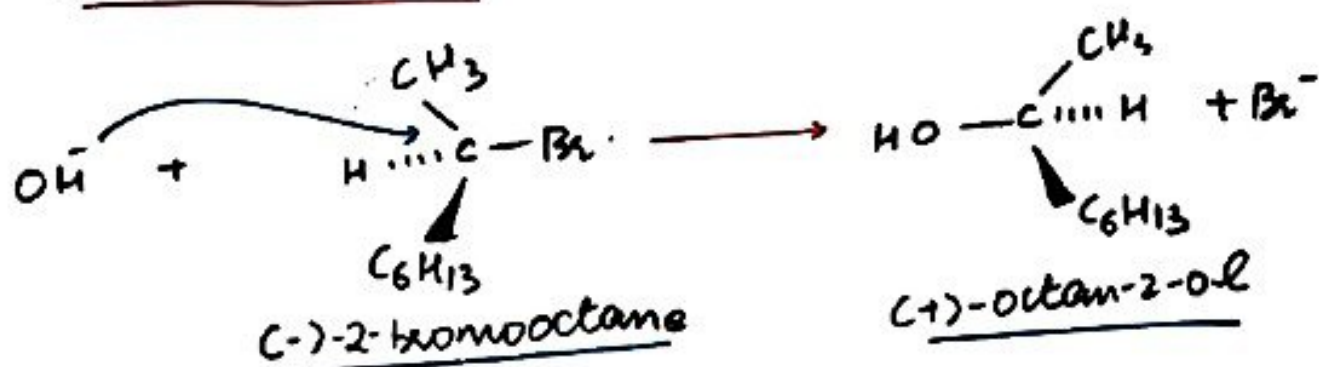
Retention → If A is the only product. Then this process is called retention.

Inversion → If B is the only product in reaction, then this process is inversion.

Racemisation → If A and B are 50%, then its process is called racemisation and mixture of A + B → Racemic mixture.

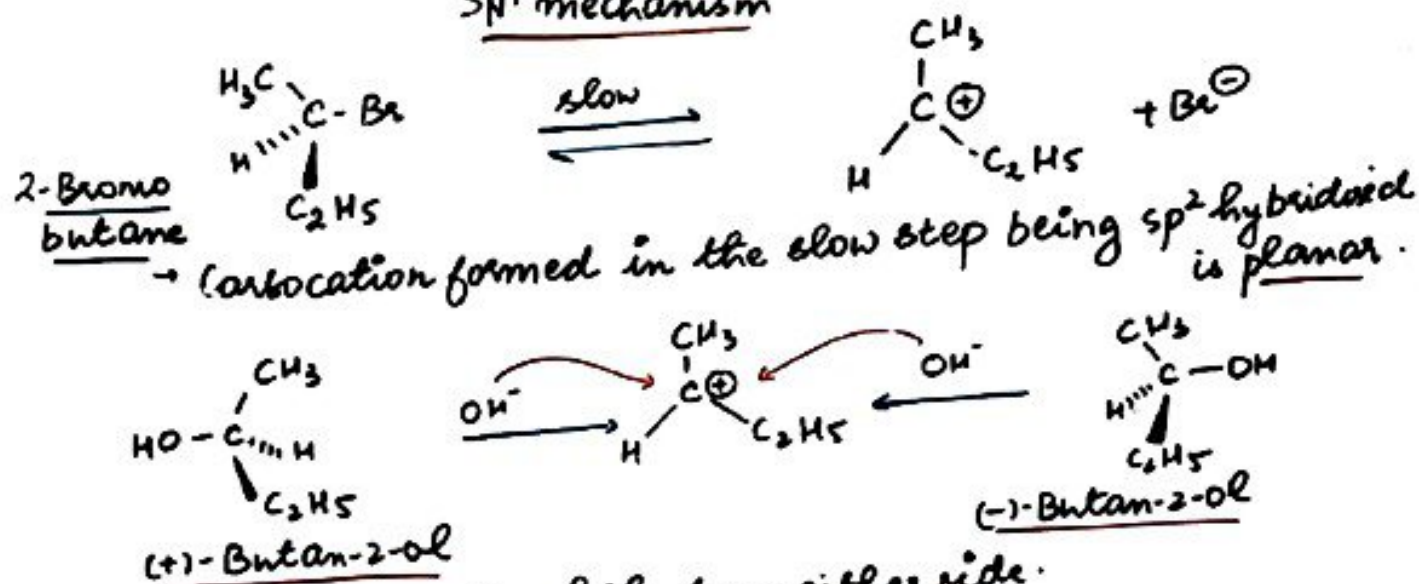
Optically active alkyl halides

S_N2 mechanism



- Inversion of configuration
- Nucleophile attaches itself on the side opposite to the one where halogen atom is present
- OH group occupying the position opposite to what bromide has occupied.

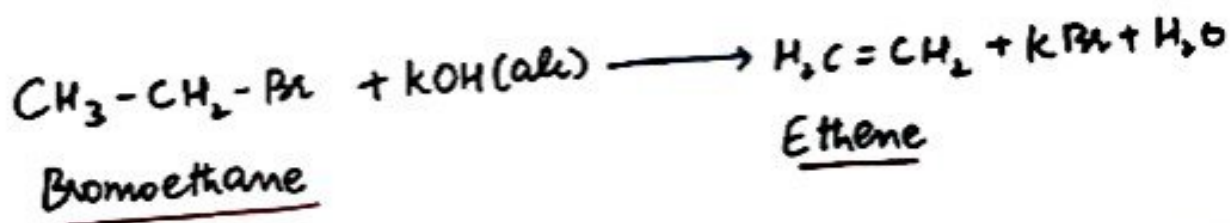
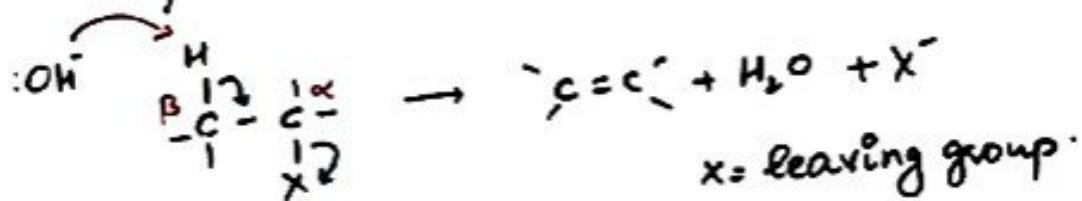
S_N1 mechanism



- Attack of nucleophile from either side.
- Products obtained have opposite configuration.
- eg Hydrolysis of 2-bromobutane result in (±)-butan-2-ol → Racemisation

2. Elimination reactions

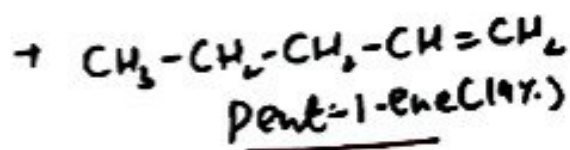
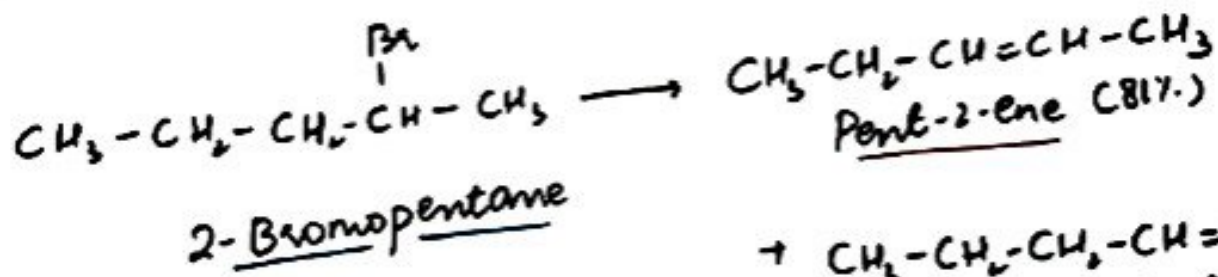
- When haloalkane with β -hydrogen atom are heated with alcoholic KOH, they undergo dehydrohalogenation to form alkene.
- These reactions are called β -elimination because the hydrogen atom present at β -position of the halogen is removed.



- If there is formation of more than one alkene due to presence of more than one β hydrogen atom.

Saytzeff rule

- In dehydrohalogenation reaction, the preferred product is that alkene which has the greater number of alkyl group attached to the doubly bonded carbon atom.

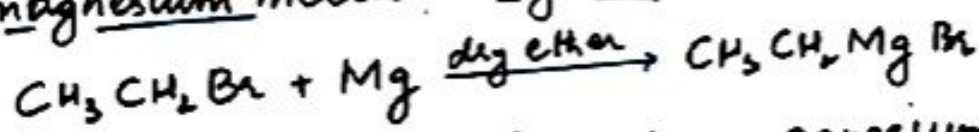


3. Reaction with metal

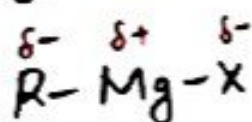
→ Most organic chlorides, bromides and iodides react with certain metal to give compound containing carbon-metal bonds. Such compounds are known as organometallic compounds.

Grignard reagent (RMgX)

→ This reagent is obtained by the action of haloalkane with magnesium metal in dry ether.

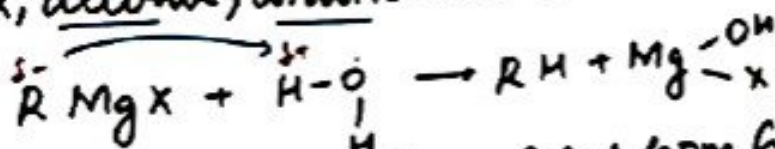


→ In Grignard reagent the carbon-magnesium bond is covalent highly polar, the magnesium halide bond is ionic.



→ Grignard reagent are highly reactive and react with any source of proton to form hydrocarbon.

→ Even water, alcohol, amine are converted to corresponding hydrocarbon.

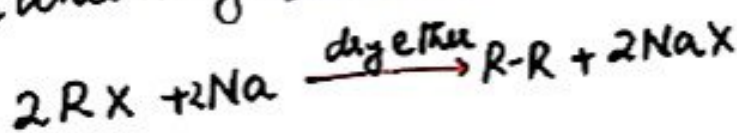


→ Even traces of moisture is avoided from Grignard reagent.

→ On other hand, it is one of method of converting halide to hydrocarbon.

Wurtz reaction

→ Alkyl halide react with sodium in dry ether to give hydrocarbon containing double number of carbon atoms present in halide.



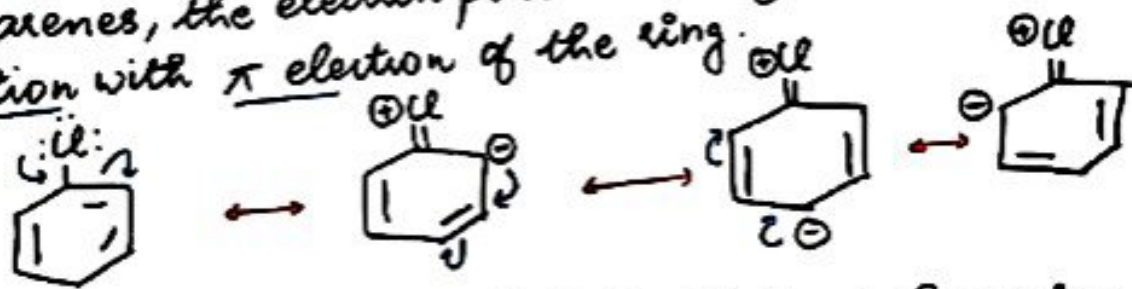
Reaction of haloarenes

1. Nucleophilic substitution

Aryl halides are less reactive toward nucleophilic substitution due to following reasons

① Resonance effect

In haloarenes, the electron pair on halogen atoms are in conjugation with π electron of the ring.



→ C-Cl bond acquire a partial double bond character due to resonance

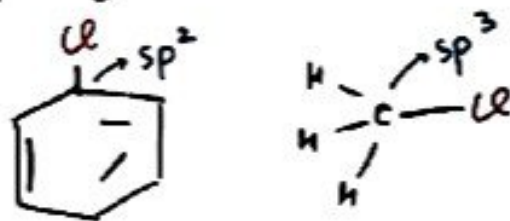
→ As a result, the bond cleavage is difficult in haloarenes

② Difference in hybridisation of carbon atom in C-X bond

→ In haloalkane, the carbon atom bearing halogen is sp^3 hybridised while halogen bearing carbon atom is sp^2 hybridised in haloarene

→ sp^2 hybrid orbital is smaller in size due to greater s character as compared with sp^3 hybrid orbital

→ As a result bond formed by overlap of sp^2 hybrid orbital is shorter in size and stronger than the bond formed by overlap of sp^3 hybrid orbital



Instability of Phenyl cation

- In case of haloarenes, phenyl cation formed as res of self ionisation will not be stabilized by resonance.
- S_N1 mechanism is ruled out

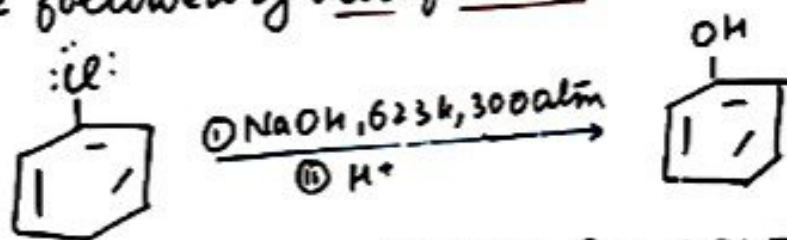


Repulsive interaction

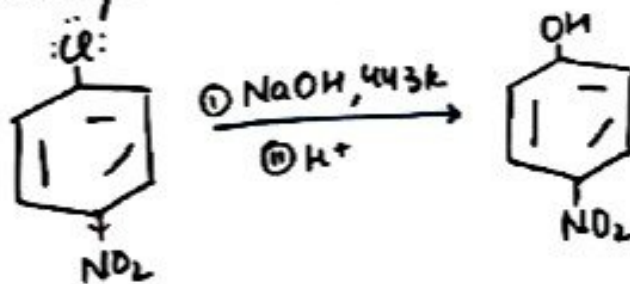
haloarenes have high electron density due to presence of localized pi electrons.
It is less likely for electron rich nucleophile to approach electron rich haloarene due to repulsive interaction.

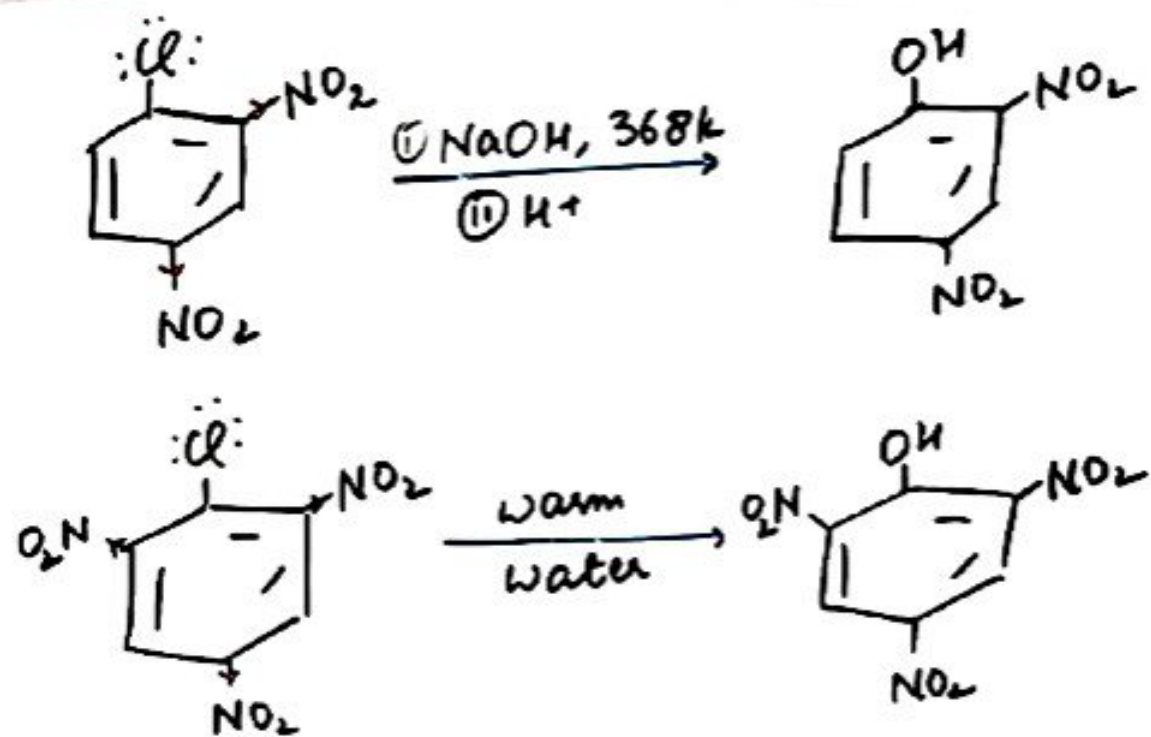
Replacement by Hydroxyl group

Chlorobenzene is converted into phenol by heating in aqueous NaOH solution at temperature of 623K and a pressure of 300 atmosphere followed by acidification.



The presence of an electron withdrawing group (-NO₂) at ortho and para position increases the reactivity of haloarene.

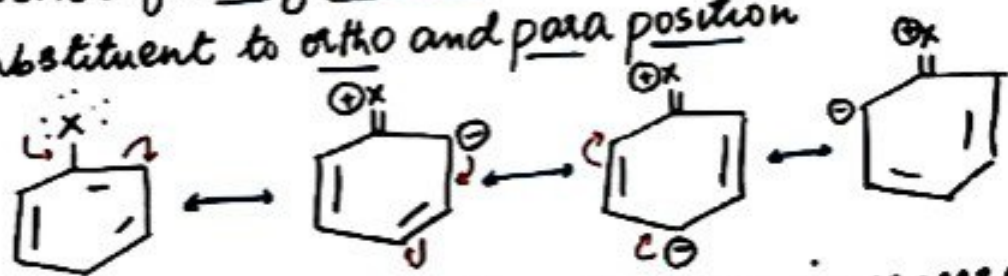




→ No effect of reactivity of haloarene observed by presence of electron withdrawing group at meta position

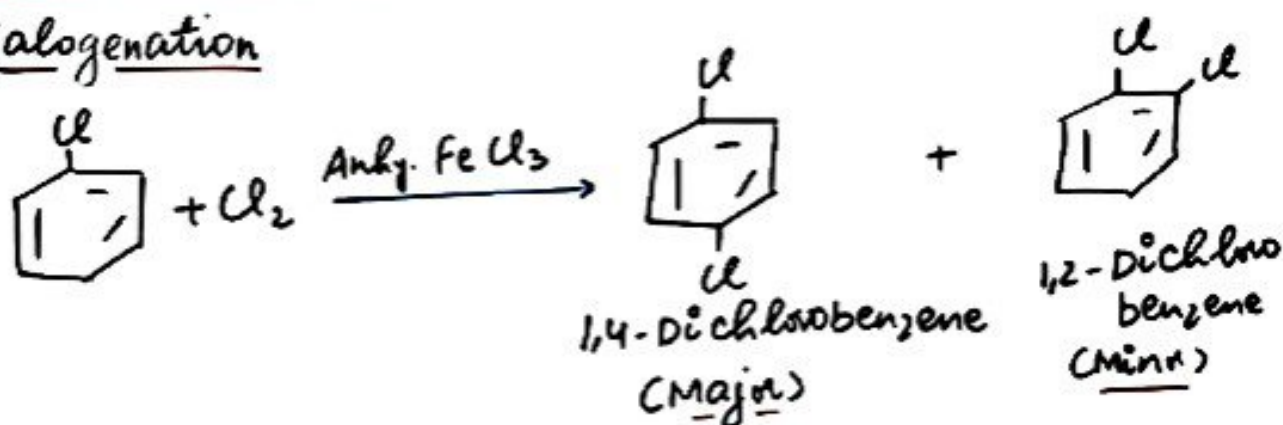
2. Electrophilic substitution reaction

- Aryl halide undergo electrophilic substitution
- The presence of halogen atom in the ring direct the incoming substituent to ortho and para position

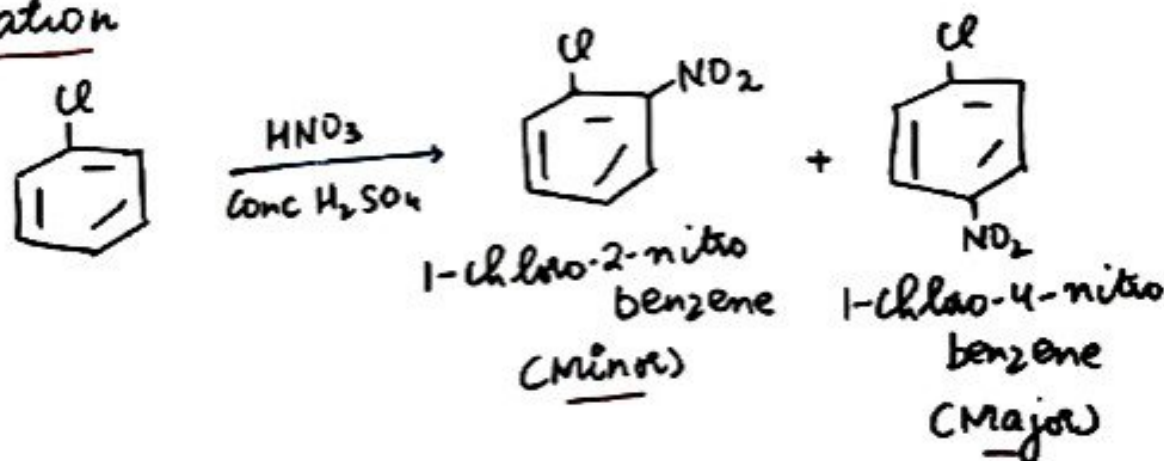


- Due to resonance, the electron density increases more at ortho and para position than at meta position
- Aryl halide are less reactive than benzene toward electrophilic substitution reaction
- Electron withdrawing inductive effect (-I effect) decreases electron density at benzene ring and ring get deactivated as compared to benzene

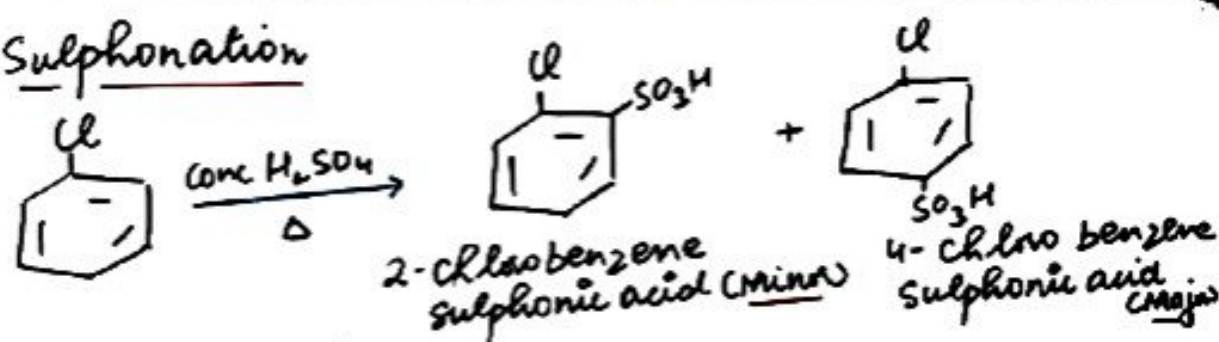
① Halogenation



② Nitration

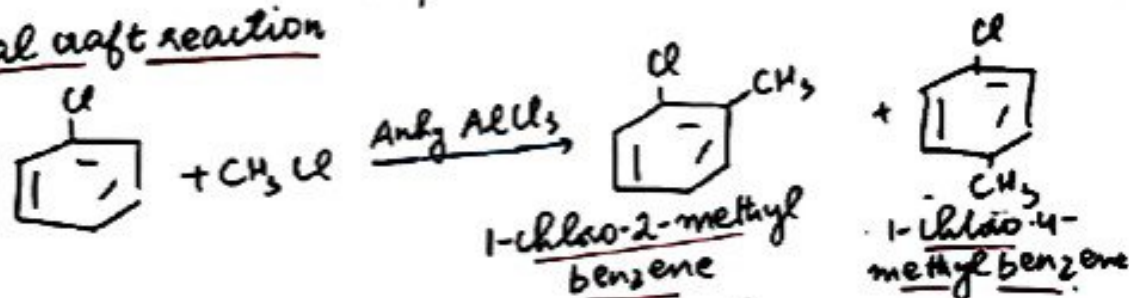


3, Sulphonation

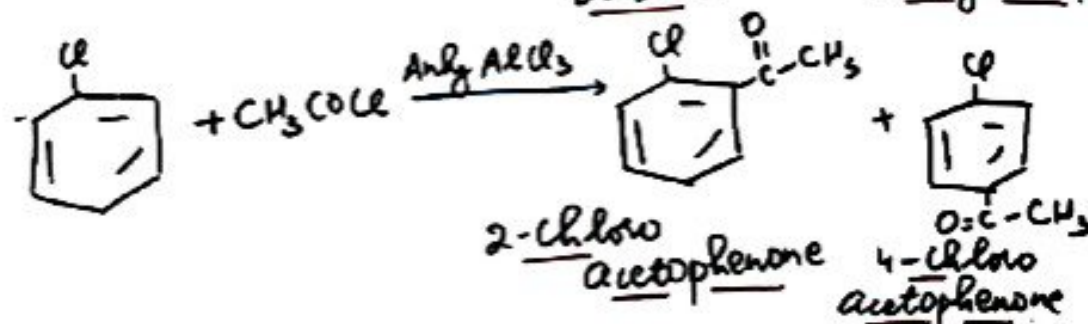


4, Friedel-Crafts reaction

Alkylation



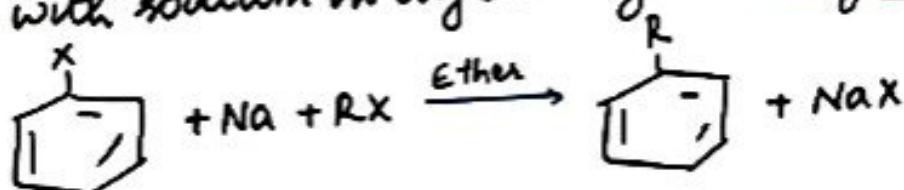
Acylation



3. Reaction with metal

Wurtz Fittig reaction

- A mixture of an alkyl halide and aryl halide
- Treated with sodium in dry ether gives alkylarene



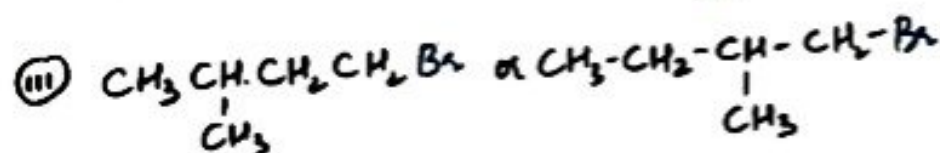
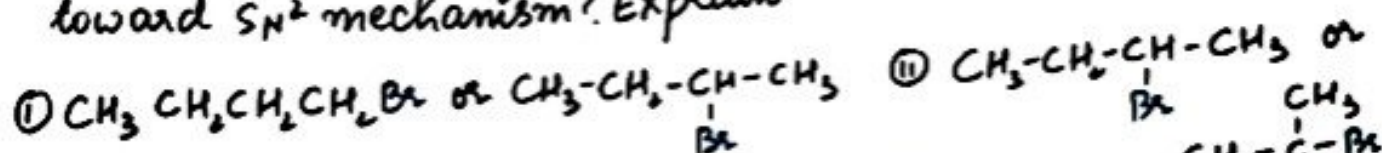
Fittig reaction

- Aryl halides treated with sodium in dry ether, in which two aryl groups are joined together



Intext Question

10.7 Which alkyl halide from the following pair react more toward $\text{S}_\text{N}2$ mechanism? Explain

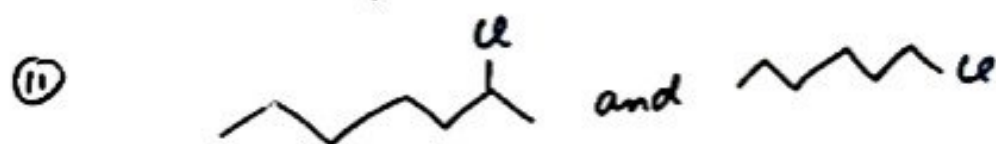
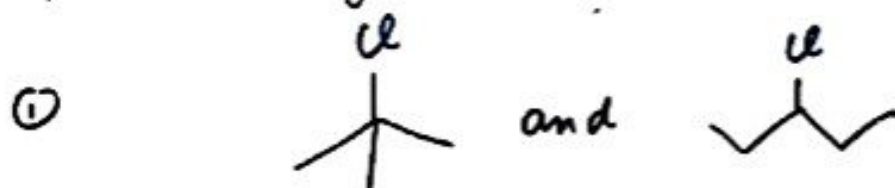


Ans ① $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ - Primary halide - less steric hindrance

② $\text{CH}_3\text{CH}_2\text{CH}(\text{Br})\text{CH}_3$ - Secondary halide - less steric hindrance
tertiary halide - less steric hindrance

③ $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{Br}$ - Presence of methyl group close to the halide group will increase steric hindrance and decrease rate of $\text{S}_\text{N}2$ mechanism

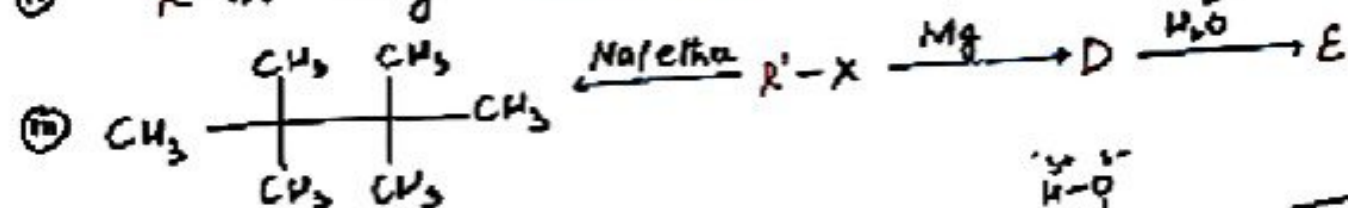
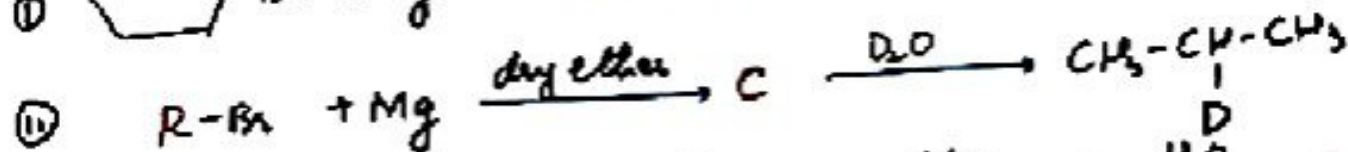
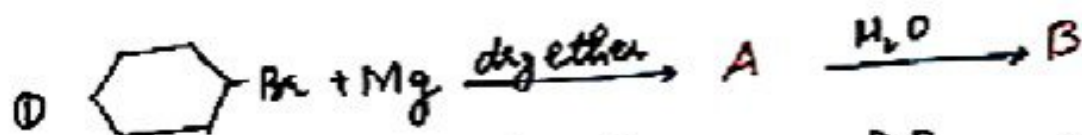
10.8 In the following pair of halogen compound, which compound undergoes faster S_N1 reaction?



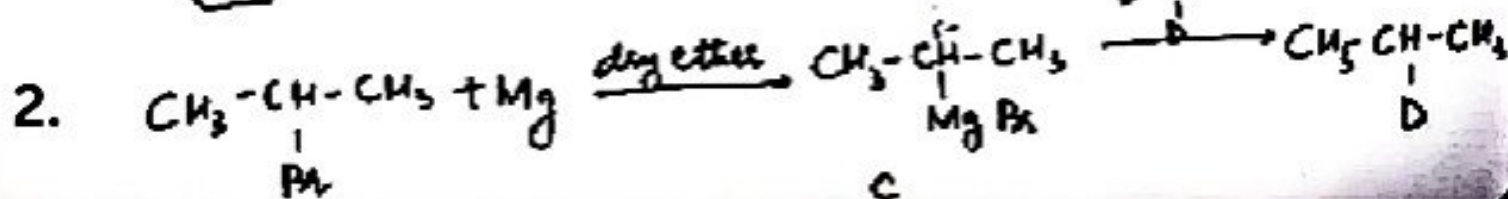
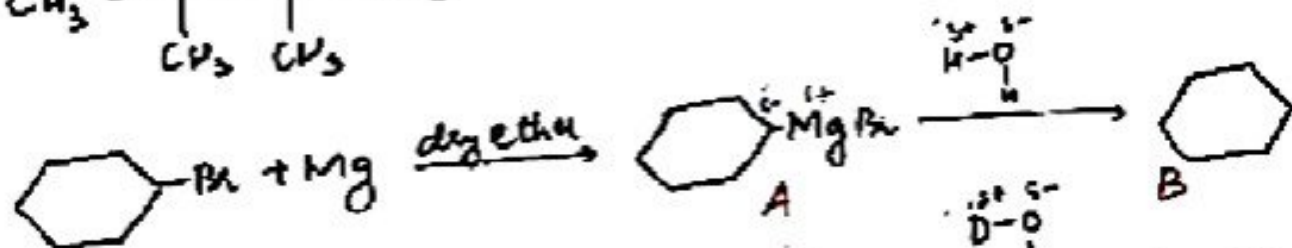
Ans (i)  - Tertiary halide react faster than secondary halide because of greater stability of 3° carbocation

(ii)  $\rightarrow$ Greater stability of 2° carbocation than 1°

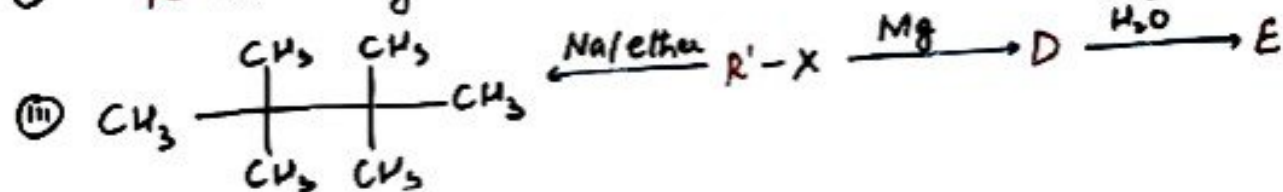
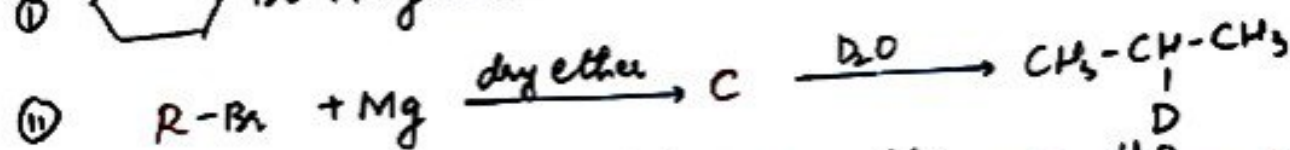
10.9 Identify A, B, C, D, E, R and R' in the following



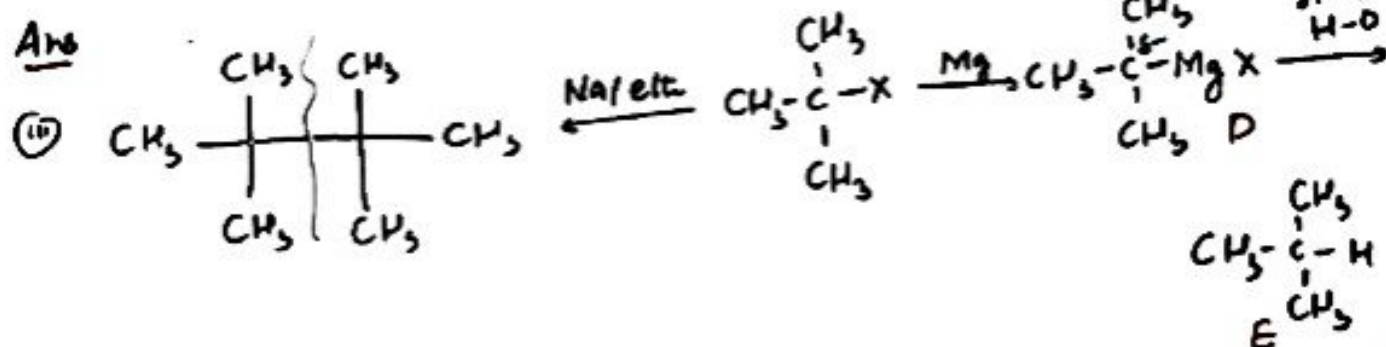
1 Ans



10.9 Identify A, B, C, D, E, R and R' in the following



Ans



Poly halogen compounds

→ Carbon compounds containing more than one halogen atom

1. Dichloromethane (Methylene chloride) (CH_2Cl_2)

Uses → Solvent

- Paint remover
- Propellant in aerosol

Harmful effect

- Harm central nervous system
- Lower level → Impaired hearing and vision
- High level → Dizziness, Nausea
→ Tingling and numbness in fingers and toes.
- Direct contact to skin
→ Burning and redness
- Direct contact with eye
→ Burn cornea

Trichloromethane (Chloroform) CHCl_3

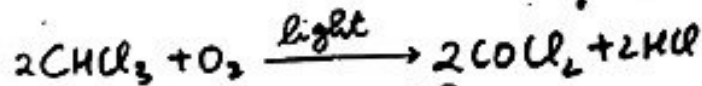
Uses

- Solvent for fat, alkaloid Iodine
- Production of Freon refrigerant R_{22}
- Once used as general anaesthetic in surgery replaced by ether
- Inhaling chloroform depresses CNS

Harmful effect

- Breathing 900ppm for short time → Dizziness, fatigue and headache

- Chronic exposure damage liver and kidney



- Extremely poisonous → Phosgene
- Stored in closed dark coloured bottle completely filled

Triiodomethane (Iodoform) CHI_3

- Earlier used as antiseptic due to liberation of free iodine.
- Due to objectionable smell, it is replaced by other formulation containing iodine.

Tetrachloromethane (Carbon tetrachloride) CCl_4

Uses

manufacture of refrigerant and propellant for aerosol can.

edstock → In Manufacture of chlorofluorocarbon

Solvent use

Unit 1960 → Cleaning fluid,

In industry → degreasing agent

Home → spot remover

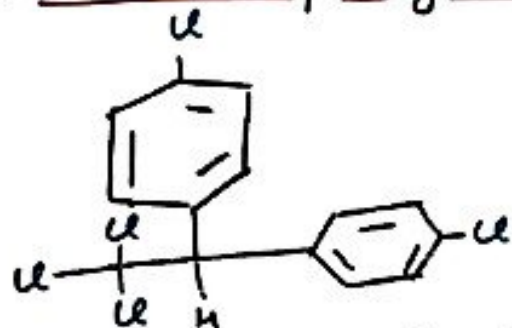
Harmful effect

- Exposure → liver cancer
- Common effect → dizziness, li headache, nausea and vomiting
- Permanent damage to nerve
- Severe cases → stupor, coma, unconsciousness or death
- Heart beat irregular
- Deplete ozone layer.

Freons

- Chlorofluoro carbon compounds of ethane and methane
- Extremely stable, unreactive, non toxic, non corrosive and easily liquefiable gas.
- Freon 12 (CCl_2F_2) - most common industries
- Manufacture from tetrachloromethane by Swarts reaction
- Aerosol propellant, refrigeration and air conditioning.
- Diffuse unchanged into stratosphere
- Initial chain reaction with ozone
- Deplete ozone layer.

P,p'-Dichlorodiphenyl trichloroethane (DDT)



Preparation → Condensation of chloral with chlorobenzene in H_2SO_4

Properties → Effective against mosquito that spread malaria and lice that carry typhus

Harmful effect → Many species of insect developed resistance
→ High toxicity toward fish
→ Stable and fat soluble → not metabolised by animal
→ Deposit and stored in fatty tissue
→ Banned in United State in 1973